

TMCG IdeaSprint 4.0 Portal

Product Requirements & Functional Specification

Purpose: This document defines the functional requirements for the TMCG IdeaSprint 4.0 Management Portal. It describes **what the platform should do**, **who can perform each action**, and **what business rules should be enforced**. It intentionally avoids implementation details so that it can be directly consumed by an AI coding agent or development team.

Event-related information in this document is based on the official IdeaSprint 4.0 event document where applicable.

1. Project Overview

The TMCG IdeaSprint 4.0 Portal is a centralized web platform that manages the complete lifecycle of the IdeaSprint 4.0 intra-campus innovation event conducted in GITAM University Visakhapatnam.

The portal should eliminate manual management by digitizing registrations, authentication, participant management, attendance, problem statement selection, NOC handling, food coupon tracking, approvals, and event administration.

- The TMCG IdeaSprint 4.0 Portal is the official event management platform for the **Visakhapatnam Campus**. The IdeaSprint 4.0 is jointly organized by **TMCG** and **Meta Developer Communities (MDC) GITAM Visakhapatnam**. It manages the complete lifecycle of the campus-level event, from registrations to participant exit.

The system should remain role-based throughout the application.

2. Event Overview

IdeaSprint 4.0 consists of two phases.

Phase 1 - Campus Level

Conducted independently at:

- Visakhapatnam
- Hyderabad
- Bengaluru

Every registered participant participates in:

Round 1

- Naukri Assessment - 100 Minutes

Round 2

- Build Hackathon - 18 hr

Participation in **both rounds is mandatory** for all registered participants. The campus-level rounds are non-eliminary and shortlisted teams progress to the Grand Finale based on their overall performance.

Phase 2 - Grand Finale

Top-performing teams from all three campuses qualify for the Grand Finale.

Cash prizes:

- ₹15,000
- ₹10,000
- ₹5,000

These prizes are awarded only during the Grand Finale and are common across all campuses.

The portal consists of the following major modules:

- Public Website
- Registration
- Authentication
- Team Management
- Participant Management
- Problem Statement Management
- NOC Management
- Attendance Management
- Food Coupon Management
- Exit Form Management
- Dashboard
- Approval Workflow
- Notifications
- Admin Configuration

Each module is protected by role-based permissions.

4. User Roles

The platform supports four user roles.

4.1 Super Admin

Has unrestricted access to the entire platform.

Responsible for overall administration and system configuration.

4.2 SPOC

Campus-level administrator.

Can perform administrative actions only for the teams assigned to them.

4.3 Team Lead

Leader of a registered team.

Responsible for managing the team's information and submissions.

4.4 Member

Regular participant belonging to a registered team.

Has limited permissions and can only access features relevant to themselves.

5. Public Website

The public website is accessible without login.

Its purpose is to provide information about IdeaSprint 4.0 and allow new teams to register.

The homepage should contain:

- Event title: **TMCG IdeaSprint 4.0**
 - Hero section
 - Vizag campus image
 - Login button
 - Register button
 - Domains section
 - Previous year event gallery
 - Event timeline
 - Prize section
 - Judges Photos
 - Frequently Asked Questions (FAQs)
 - Important Instructions
 - Contact section
 - Footer
-

6. Homepage Content

The homepage should prominently display:

Event Title

TMCG IdeaSprint 4.0

Hero Section

Contains:

- Event title
 - Vizag campus image
 - Login button
 - Register button
-

Domains Section

Display the list of domains provided by the organizers.

The domain list should be configurable by the administrator.

Gallery

Display previous-year event photographs.

Images should be manageable from the admin dashboard.

Event Timeline

Display the official event timeline.

The timeline content should be configurable from the Admin Configuration module.

Prize Section

Display:

- ₹15,000
- ₹10,000
- ₹5,000

Mention that prizes are awarded at the Grand Finale.

Registration Fee

Display:

No Registration Fee

Important Instructions

Display all event instructions.

Instructions should be editable from the Admin Configuration module.

NOC Notice

The homepage should prominently display:

NOC Submission is COMPULSORY for every participant.

This notice should remain highly visible throughout the registration period.

FAQs

Display organizer-provided FAQs.

Editable from Admin Configuration.

Footer

Footer should contain:

- 3 TMCG contact persons
- 1 MDC contact person

The following should remain configurable:

- Name
- Designation
- Phone Number
- Email Address

7. Registration Workflow

Registration is team-based.

Only one participant registers the team.

That participant automatically becomes the Team Lead.

Registration Flow

Click Register



Display Registration Guidelines



User acknowledges guidelines



Basic Team Details



Member Details



Validation



Team Registration Complete



Google Login



Dashboard Access

8. Registration Guidelines

Before registration begins:

The system should display the official registration guidelines.

The user must acknowledge the guidelines before proceeding.

Registration cannot continue unless acknowledgement is provided.

The guideline content should be editable from Admin Configuration.

9. Step 1 – Basic Team Details

The Team Lead provides:

- Team Name
- Domain
- No of Members

The system should not ask for any additional information at this stage.

10. Step 2 – Team Member Details

After Step 1 is completed,

The Team Lead proceeds to the participant details page.

The Team Lead provides his information & participant information for every team member.

Required information: All are mandatory

- Name
- Reg No
- GITAM Mail ID
- Phone Number
- Year of study
- School

- Department
- Branch
- Gender
- Stay

The Team Lead is also one of the participants.

11. Team Size Rules

As per the official event document:

- Minimum members: 3
- Maximum members: 4

The Team Lead counts as one team member.

12. Registration Validations

The system must validate the following before completing registration.

Team Name

Must be unique.

Duplicate team names are not allowed.

University Email

Must be unique.

The same university email cannot exist in multiple teams.

Registration Number

Must be unique.

The same participant cannot be registered twice.

Mobile Number

Must be unique.

Duplicate mobile numbers should not be accepted.

Existing Participant Validation

If a participant is already registered in another team,
the registration should fail.

Validation should occur using:

- Registration Number
- University Email

The system should clearly indicate that the participant already belongs to another team.

Google Authentication Domain

Only university email IDs are allowed.

Accepted domain:

@student.gitam.edu or gitam.in

Any other domain should be rejected.

13. Team Registration Completion

Registration is considered successful only when:

- All validations pass
- Team size is valid

- Every participant's mandatory information is provided
- No duplicate participant exists
- Team name is unique

Once registration is completed:

- A unique Team ID is generated.
 - Every participant receives a unique User ID.
-

14. Team ID

Every team should receive a unique Team ID.

The format is configurable.

The generation logic should be handled by the system.

15. User ID Generation

Every participant receives a unique User ID.

Examples:

VSP1001
VSP1002
VSP1003

The numbering should increment automatically.

16. Authentication

Authentication is available only after successful registration.

Users who have not registered cannot access protected areas of the portal.

Authentication uses:

- Google Authentication

- Supabase Authentication

Authentication is restricted to university Google accounts only.

17. Login Flow

Login



Google Authentication



Verify University Email



Fetch User



Identify Role



Redirect to Dashboard

18. Dashboard Routing

After successful authentication:

If Super Admin

→ Super Admin Dashboard

If SPOC

→ SPOC Dashboard

If Team Lead

→ Team Dashboard

If Member

→ Member Dashboard

No intermediate landing page is required because users have already acknowledged the registration guidelines before creating their team.

19. Session Management

Authenticated users should remain logged in until:

- They explicitly log out, or
- Their session expires(15 Mins) according to the authentication provider's configuration.

Users attempting to access protected routes without authentication should be redirected to the login page.

20. Business Rules (Part 1)

- Registration is team-based.
 - Only one Team Lead exists per team.
 - Team Lead registers the entire team.
 - Registration guidelines must be acknowledged before registration.
 - Team names must be unique.
 - University emails must be unique.
 - Registration numbers must be unique.
 - Mobile numbers must be unique.
 - A participant cannot belong to multiple teams.
 - Only [@student.gitam.edu](mailto:student.gitam.edu) or gitam.in accounts can authenticate.
 - Registration must be completed before login.
 - User IDs are campus-specific.
 - Team IDs are system-generated.
 - Dashboard access is based on user role.
 - Homepage content should be configurable through the Admin Configuration module wherever applicable.
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21. Team Management Module

The Team Management module is the central workspace for every registered team.

It stores all information related to:

- Team
- Team Lead
- Members
- Problem Statement
- NOCs
- Attendance
- Food Coupons
- Exit Status

The functionality available within this module depends on the user's role.

22. Team Profile

Every registered team shall have a Team Profile.

The Team Profile acts as the single source of truth for all team-related information.

It should contain:

Basic Details

- Team ID
 - Team Name
 - Registration Status
 - Current Team Status
-

Team Lead

Display:

- Name
- Reg No
- GITAM Mail ID
- Phone Number
- Year of study

- School
 - Department
 - Branch
 - Gender
 - Stay
 - User ID
-

Team Members

Display every participant with:

- User ID
 - Name
 - Reg No
 - GITAM Mail ID
 - Phone Number
 - Year of study
 - School
 - Department
 - Branch
 - Gender
 - Stay
-

Event Information

Display:

- Selected Problem Statement
 - NOC Status
 - Attendance Summary
 - Food Coupon Status
 - Exit Status
-

23. Team Status

Every team shall have one status.

Possible statuses include:

- Registered

- Active
- Pending Approval
- Qualified for Grand Finale
- Exited

The system should automatically update the team status whenever applicable.

24. Team Editing

Only the Team Lead can edit team information like Participant details & his details etc and all

User ID should never be editable.

Team ID should never be editable.

25. Approval Workflow for Team Changes

Whenever the Team Lead edits team information:

The changes should **not** be applied immediately.

Instead,

the request should be marked:

Pending Approval

Workflow

Team Lead edits information

↓

Submit Request



Pending Approval



SPOC / Super Admin Review



Approve



Apply Changes

OR

Reject



Keep Previous Information

26. Pending Changes

While awaiting approval:

The system should display:

Current Information



Requested Changes



Approval Status

The Team Lead should not be able to submit another modification request until the existing one is resolved.

27. Member Permissions

Members can only:

- View Team Information
- View Team Members
- View Problem Statement
- View Attendance
- View Food Coupon Status
- Upload NOC
- View their uploaded NOC
- View Exit Status

Members cannot:

- Edit Team
 - Delete Team
 - Delete NOC
 - Edit NOC
 - Select Problem Statement
 - Edit Problem Statement
 - Approve Requests
-

28. Team Lead Permissions

The Team Lead can:

- View Team
- Edit Team
- Submit Team Edit Request
- Upload NOC
- View NOC
- Edit NOC
- Delete NOC
- Select Problem Statement
- Change Problem Statement
- View Attendance
- View Food Coupons
- Upload Exit Form

The Team Lead cannot:

- Approve Requests
 - Assign SPOCs
 - Modify other teams
-

29. Team Dashboard

The Team Dashboard should provide quick access to:

- Team Profile
- Members
- Problem Statement
- Attendance
- Food Coupons
- NOC
- Exit Form
- Notifications

This should serve as the primary workspace for Team Leads.

30. Problem Statement Module

The Problem Statement module manages the complete lifecycle of problem statement release and selection.

31. Problem Statement Visibility

Problem Statements should **not** be visible immediately.

The administrator specifies:

- Release Date
- Release Time

Before the configured release time,

participants should not be able to view any Problem Statements.

32. Problem Statement Release

Spreadsheet link will be given for the members to view the problem statements.

33. Problem Statement Selection

The Team Lead selects the desired Problem Statement.

Selection is performed by entering the

Problem Statement Number

rather than browsing and selecting directly.

The system validates the entered code.

34. Selection Time Window

Problem Statement selection is allowed only within a configured time window.

The administrator configures:

- Selection Start
- Selection End

Outside this duration,
selection is disabled.

35. Editing Problem Statement

Before the deadline, the Team Lead may change the selected Problem Statement any number of times.

Only the latest selection should be considered final.

After the deadline,
editing is disabled.

36. Super Admin & SPOC Extension

If a team fails to select a Problem Statement before the deadline,

the Super Admin & SPOC should be able to extend the selection window for that specific team.

The extension should require:

- Duration
- Reason (optional, configurable)

The extension affects only the selected team.

Other teams continue following the original deadline.

37. Problem Statement Capacity

There is **no limit** on the number of teams selecting a particular Problem Statement.

Multiple teams may choose the same Problem Statement.

38. Problem Statement History

The system should maintain:

- Initial Selection
- Every Modification
- Final Selection
- Time of Selection

This history is for administrative purposes.

39. NOC Module

NOC submission is mandatory for every participant.

The portal should prominently communicate this requirement.

40. Individual NOCs

Every participant has an individual NOC.

NOCs are managed separately.

There is no shared team NOC.

41. Team Lead NOC Permissions

The Team Lead can:

- Upload
- View
- Replace
- Delete

NOCs belonging to every member of the team.

42. Member NOC Permissions

Members can:

- Upload their own NOC
- View their own NOC

Members cannot:

- Edit
- Replace
- Delete

their uploaded NOC.

If a Team Lead deletes a member's NOC,

the member should again be able to upload it.

43. NOC Status

Each participant should have an individual NOC status.

Possible values:

- Not Uploaded
 - Uploaded
 - Verified (*if verification is introduced later*)
 - Missing
-

44. Team NOC Summary

The Team Dashboard should display:

Member 1

Uploaded

Member 2

Uploaded

Member 3

Missing

Member 4

Uploaded

This enables the Team Lead to quickly identify pending submissions.

45. NOC Deadline

The NOC submission deadline should be configurable from the Admin Configuration module.

After the deadline,

upload permissions should follow the administrator's configured policy.

(No behavior beyond configurability is specified.)

46. Notifications

The Team should receive notifications when:

- NOC Uploaded
- NOC Deleted
- Problem Statement Released
- Problem Statement Changed

- Problem Statement Deadline Extended
 - Team Changes Approved
 - Team Changes Rejected
-

47. Validation Rules

Team Name

Unique.

Registration Number

Unique across the platform.

University Email

Unique across the platform.

Phone Number

Unique across the platform.

Team Membership

A participant can belong to only one team.

Problem Statement

Must exist.

NOC Upload

Accepted only for registered participants.

48. Business Rules

- Team Lead is responsible for maintaining team information.
 - Members cannot edit team information.
 - Team edits require approval.
 - Problem Statements are released only after the configured release time.
 - Problem Statements can be changed multiple times before the deadline.
 - SPOCs can extend the selection deadline for specific teams only.
 - There is no Problem Statement selection limit.
 - Every participant requires an individual NOC.
 - Members can upload and view only their own NOC.
 - Team Leads can manage all NOCs belonging to their team.
 - NOC submission is mandatory.
 - Every modification should be recorded in system history where applicable.
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49. Attendance Module

The Attendance Module is responsible for recording participant attendance throughout the event.

Attendance is maintained **per participant**, while being grouped under their respective teams.

This allows organizers to monitor both team-level and participant-level attendance.

50. Attendance Sessions

Attendance is session-based.

The number of attendance sessions is configurable from the Admin Configuration Module.

Example:

- Attendance Session 1
- Attendance Session 2
- Attendance Session 3

- Attendance Session 4

The system should not hardcode the number of sessions.

51. Attendance Record

Each attendance record should contain:

- Attendance Session
 - Participant
 - Team
 - Attendance Status
 - Recorded By
 - Recorded Time
-

52. Attendance Status

Possible values:

- Present
- Absent

No other attendance statuses are currently required.

53. Attendance Entry

Only the assigned SPOC and Super Admin can record attendance.

Attendance should be entered participant-wise.

Example:

Team Alpha

✓ Member 1

✓ Member 2

✗ Member 3

✓ Member 4

The system should automatically associate participants with their teams.

54. Attendance Modification

Attendance may require correction.

Only:

- Super Admin
- Assigned SPOC

can modify an attendance record.

The system should maintain an audit history of attendance modifications.

55. Attendance History

Every attendance modification should record:

- Previous Status
- New Status
- Modified By
- Modified Time

This history should only be visible to administrators.

56. Attendance Dashboard

Attendance Dashboard should display:

Team Summary

- Team Name
 - Present Count
 - Absent Count
-

Participant Summary

For every participant:

- Name
 - User ID
 - Attendance Status
-

Session Summary

Per session:

- Total Present
 - Total Absent
 - Attendance Percentage
-

57. Food Coupon Module

The Food Coupon Module tracks meal redemption during the event.

It is **not** responsible for generating coupons.

Its purpose is to record whether a participant has redeemed their meal.

58. Meals

The system should support recording:

- Lunch
- Dinner

Breakfast is **not provided** during the event and should not appear as a redeemable meal.

59. Coupon Status

Each meal has two possible states:

- Redeemed
- Not Redeemed

No additional workflow is required.

60. Coupon Recording

Only:

- Assigned SPOC
- Super Admin

can mark a meal as Redeemed or Not Redeemed.

Participants cannot change their own coupon status.

61. Coupon Dashboard

For every participant display:

Lunch

- Redeemed / Not Redeemed

Dinner

- Redeemed / Not Redeemed
-

For every team display:

Total Lunch Redeemed

Total Dinner Redeemed

62. Attendance + Food Integration

Attendance and Food Coupons should be manageable from the same administrative workspace.

Organizers should not need to switch modules while managing event sessions.

The attendance view should allow SPOCs to simultaneously record:

- Attendance
 - Lunch Redemption
 - Dinner Redemption
-

63. Exit Form Module

The Exit Form Module is used after the event concludes.

The physical Exit Form is managed outside the portal.

However, the signed form must be uploaded into the portal.

64. Exit Form Upload

Only the Team Lead can upload the team's Exit Form.

Members cannot upload or modify Exit Forms.

65. Exit Form Requirements

The uploaded Exit Form should already contain the required physical signatures before upload.

The portal is **not** responsible for collecting digital signatures.

66. Exit Status

Each team should have an Exit Status.

Possible values:

- Not Submitted
- Submitted
- Verified (*if verification is introduced later*)
- Exited

Uploading the Exit Form should update the team's status accordingly.

67. Exit Dashboard

The Team Dashboard should display:

- Exit Form Uploaded
- Exit Status

The Administrative Dashboard should display:

- Teams Pending Exit
 - Teams Completed Exit
-

68. Super Admin

The Super Admin has unrestricted access to the platform.

This role is responsible for complete administration of the event.

69. Super Admin Permissions

The Super Admin can:

User Management

- View all users
 - Manage all users
 - Assign roles
 - Change roles
-

Team Management

- View all teams
 - Edit any team
 - Delete teams (*if required by organizers*)
 - Approve team modifications
-

SPOC Management

- Create SPOCs
 - Assign SPOCs
 - Reassign SPOCs
 - Remove SPOCs
-

Attendance

- View attendance
 - Record attendance
 - Edit attendance
-

Food Coupons

- Mark Redeemed
 - Mark Not Redeemed
-

Problem Statements

- Create
- Edit
- Delete
- Release
- Hide

NOCs

- View every NOC
 - Manage every NOC
-

Exit Forms

- View all Exit Forms
 - Manage Exit Forms
-

Dashboard

Access every dashboard.

Reports

Generate reports for:

- Teams
 - Participants
 - Attendance
 - Food Coupons
 - NOCs
 - Exit Status
-

Configuration

Manage all configurable settings.

70. SPOC

Every SPOC is assigned specific teams.

A SPOC has administrative permissions only for those assigned teams.

71. SPOC Permissions

A SPOC can:

- View assigned teams
- Manage assigned teams
- Approve Team Edit Requests
- Record Attendance
- Modify Attendance
- Record Food Redemption
- View NOCs
- Manage NOCs
- Extend Problem Statement selection deadline
- View Exit Forms
- View Notifications

A SPOC cannot:

- Manage teams belonging to another SPOC
 - Modify system-wide configuration
 - Assign other SPOCs
 - Create administrators
-

72. Team Assignment

Every registered team should be assigned to a SPOC.

The assignment is performed only by the Super Admin.

A team can belong to only one SPOC at any point in time.

The Super Admin may reassign a team whenever required.

73. Approval Dashboard

Both:

- Super Admin
- SPOC

should have a common approval queue.

Pending requests include:

- Team Detail Changes

Future approval workflows can also be added to this queue.

74. Approval Workflow

Team Lead submits request

↓

Pending

↓

SPOC / Super Admin reviews

↓

Approve

↓

Apply Changes

OR

Reject

↓

Keep Existing Data

↓

Notify Team Lead

75. Notifications

Administrative users should receive notifications for:

- New Team Registration
 - Pending Approval
 - NOC Upload
 - Exit Form Upload
 - Attendance Completion
 - Problem Statement Deadline Extension Requests (*if introduced later*)
-

76. Dashboard Metrics

Super Admin Dashboard

Display:

- Total Registrations
 - Total Teams
 - List of Teams
 - List of Participants
 - Pending Approvals
 - NOC Completion
 - Exit Completion(IF ANY)
 - Food Redemption Summary
-

SPOC Dashboard

Display:

- Assigned Teams
 - Pending Approvals
 - Today's Attendance
 - Missing NOCs
 - Exit Forms Pending
 - Food Redemption Status
-

77. Audit History

Administrative actions should be logged.

Examples:

- Attendance Edited
- Team Edited
- Team Approved
- Team Rejected
- NOC Deleted
- Problem Statement Changed
- SPOC Assignment Changed

Each log should contain:

- User
 - Action
 - Timestamp
 - Previous Value (where applicable)
 - New Value (where applicable)
-

78. Business Rules

- Attendance is participant-wise.
 - Attendance is grouped under teams.
 - Attendance sessions are configurable.
 - Only SPOCs and Super Admins can manage attendance.
 - Breakfast is not tracked.
 - Food Coupons only record redemption status.
 - Exit Forms are physical documents uploaded after signing.
 - Team Leads upload Exit Forms.
 - Every team belongs to one SPOC.
 - SPOCs only manage assigned teams.
 - Super Admin has unrestricted platform access.
 - Administrative actions should be auditable.
-

79. Admin Configuration Module

The Admin Configuration Module acts as the central control panel for all configurable aspects of the event.

No event-specific values should be hardcoded unless they are permanent business rules.

This allows the same portal to be reused for future editions of IdeaSprint with minimal changes.

Only the Super Admin can access this module.

80. Event Configuration

The Super Admin should be able to configure:

- Event Name
 - Event Description
 - Event Banner
 - Hero Section Content
 - Homepage Announcement
 - Registration Status (Open / Closed)
 - Registration Start Date & Time
 - Registration End Date & Time
 - Event Start Date
 - Event End Date
 - Grand Finale Information
 - Prize Information
-

81. Homepage Configuration

The homepage should be completely manageable from the Admin Dashboard.

Configurable items include:

- Hero Image
 - Hero Title
 - Event Description
 - Domain List
 - Previous Year Gallery
 - Important Instructions
 - FAQs
 - Contact Details
 - Footer Information
 - Announcement Banner
-

82. Timeline Configuration

Administrators should be able to configure:

- Round 1 Start
- Round 1 End
- Problem Statement Release
- Problem Statement Selection Start
- Problem Statement Selection End
- Build Phase
- Submission Deadline
- Evaluation Time
- Winner Announcement

The portal should automatically follow these configured timings.

83. Problem Statement Configuration

The administrator should be able to:

- Create Problem Statements
- Edit Problem Statements
- Delete Problem Statements
- Assign Problem Statement Numbers
- Release Problem Statements
- Hide Problem Statements

Each Problem Statement should contain:

- Problem Statement Number
 - Title
 - Description
 - Status (Hidden / Released)
-

84. Attendance Configuration

The administrator should be able to configure:

- Number of Attendance Sessions
- Session Names
- Session Timings
- Attendance Window

Attendance should automatically follow these settings.

85. Registration Configuration

The administrator should be able to configure:

- Registration Guidelines
 - Registration Timeline
 - Registration Status
 - Maximum Registration Capacity (*if introduced later*)
-

86. NOC Configuration

The administrator should be able to configure:

- NOC Submission Deadline
 - Accepted File Types
 - Maximum File Size
-

87. Contact Configuration

The administrator should be able to manage:

- TMCG Contacts
- MDC Contact

Each contact should include:

- Name
 - Designation
 - Phone Number
 - Email Address
-

88. Notification Configuration

The administrator should be able to enable or disable notifications for different events.

89. Notification System

Notifications should be generated automatically whenever important actions occur.

Notifications should appear inside the dashboard.

Future support for email notifications can be added without changing the existing workflow.

90. Participant Notifications

Participants should receive notifications when:

- Registration Completed
 - Team Edit Approved
 - Team Edit Rejected
 - Problem Statements Released
 - Problem Statement Changed
 - Problem Statement Deadline Extended
 - NOC Uploaded
 - NOC Deleted
 - Exit Form Uploaded
 - Event Announcements Published
-

91. Administrative Notifications

Super Admin and SPOCs should receive notifications for:

- New Team Registration
 - Pending Team Edit Approval
 - NOC Uploads
 - Exit Form Uploads
 - Attendance Completion
 - Attendance Modification
 - Problem Statement Updates
-

92. Validation Rules

The system should enforce the following validations throughout the platform.

Team

- Team Name must be unique.
 - Team ID must be unique.
-

Participant

- Registration Number must be unique.
 - University Email must be unique.
 - Mobile Number must be unique.
 - A participant cannot belong to multiple teams.
-

Authentication

- Only `@student.gitam.edu` accounts can authenticate.
 - Users cannot access dashboards before registration.
-

Problem Statement

- Selection allowed only during the configured time window.
 - Teams can modify selections until the configured deadline.
 - SPOC extensions override the deadline only for the selected team.
-

NOC

- Every participant must have an individual NOC.
 - Members cannot edit or delete uploaded NOCs.
 - Team Leads can manage all NOCs within their team.
-

Attendance

- Attendance can only be recorded by the assigned SPOC or Super Admin.
- Attendance is recorded per participant.

Exit Form

- Only Team Leads can upload Exit Forms.
 - Members cannot upload or modify Exit Forms.
-

93. Business Rules

The following rules govern the overall platform:

- Registration is team-based.
 - One Team Lead per team.
 - Team size must follow the official event rules (minimum 3, maximum 4).
 - Team names are unique.
 - Registration Numbers are unique.
 - University Emails are unique.
 - Mobile Numbers are unique.
 - Participants cannot belong to multiple teams.
 - Registration must be completed before login.
 - Google Authentication is restricted to university email IDs.
 - Team modifications require approval.
 - Attendance is participant-wise.
 - Food Coupons only record redemption.
 - Problem Statements are released according to configured timelines.
 - Problem Statement selection has no participant limit.
 - Team Leads may modify Problem Statement selections until the deadline.
 - SPOCs may extend the selection window for individual teams.
 - Every participant must submit an individual NOC.
 - Team Leads upload the team's Exit Form after collecting physical signatures.
 - Every team is assigned to exactly one SPOC.
 - Super Admin has unrestricted platform access.
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94. Suggested Database Entities

The platform should include entities for:

User

Stores:

- User ID
 - Name
 - Email
 - Mobile Number
 - Registration Number
 - Role
 - Academic Year
-

Team

Stores:

- Team ID
 - Team Name
 - Team Lead
 - SPOC
 - Team Status
-

Team Members

Maps users to teams.

Problem Statement

Stores:

- Number
 - Title
 - Description
 - Status
-

Problem Statement Selection

Stores:

- Team
 - Selected Problem Statement
 - Selection Time
 - Modification History
-

Attendance

Stores:

- Participant
 - Session
 - Status
 - Recorded By
 - Timestamp
-

Food Coupon

Stores:

- Participant
 - Lunch Status
 - Dinner Status
-

NOC

Stores:

- Participant
 - Uploaded File
 - Uploaded By
 - Upload Time
-

Exit Form

Stores:

- Team
- Uploaded File
- Uploaded By

- Upload Time
 - Exit Status
-

Notifications

Stores:

- Recipient
 - Title
 - Message
 - Read Status
 - Timestamp
-

Approval Requests

Stores:

- Request Type
 - Requested Changes
 - Requested By
 - Status
 - Reviewed By
 - Review Timestamp
-

Audit Logs

Stores:

- User
 - Action
 - Previous Value
 - New Value
 - Timestamp
-

Configuration

Stores all configurable values used throughout the application.

95. Edge Cases

The system should gracefully handle scenarios such as:

Registration

- Duplicate team name.
 - Duplicate participant.
 - Duplicate email.
 - Duplicate registration number.
 - Duplicate mobile number.
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Authentication

- Unregistered user attempts login.
 - Non-university Google account attempts login.
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Problem Statement

- Team attempts selection before release.
 - Team attempts selection after deadline.
 - Team attempts modification after deadline.
 - Team enters an invalid Problem Statement Number.
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NOC

- Member uploads duplicate file.
 - Team Lead deletes a member's NOC.
 - Member re-uploads after deletion.
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Attendance

- Attendance modified after submission.
 - Same participant marked twice for the same session.
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Exit Form

- Team uploads multiple Exit Forms.

- Team attempts upload after event closure (behavior configurable).
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96. Performance Requirements

The platform should:

- Support approximately 600–1000 concurrent users.
 - Maintain responsive performance during peak registration periods.
 - Handle simultaneous logins across all campuses.
 - Support concurrent administrative operations without data inconsistency.
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97. Security Requirements

The platform should ensure:

- Secure authentication using Google Auth via Supabase.
 - Role-based authorization.
 - Protection against unauthorized access.
 - Secure file uploads.
 - Audit logging of administrative actions.
 - Protection of participant data.
 - Restricted access to team information based on role.
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98. Future Scope

The architecture should allow future additions without requiring major structural changes.

Possible enhancements include:

- Email Notifications
- SMS Notifications
- QR-based Attendance
- QR-based Food Coupon Redemption
- Judge Portal
- Evaluation Module
- Live Leaderboard
- Certificate Generation
- Grand Finale Management

- Analytics Dashboard
 - Campus Comparison Dashboard
 - Export Reports (PDF/Excel)
 - Volunteer Module
 - Sponsor Management
 - Public Result Portal
 - API Integrations
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99. Acceptance Criteria

The platform will be considered functionally complete when:

- The public website is operational.
 - Team registration workflow is fully functional.
 - Google Authentication is restricted to university email IDs.
 - All four user roles operate according to defined permissions.
 - Team management functions correctly.
 - Problem Statement workflow follows configured timelines.
 - NOC management is complete.
 - Attendance is recorded participant-wise.
 - Food coupon redemption is tracked.
 - Exit Form uploads function correctly.
 - Approval workflow operates correctly.
 - Admin Configuration controls all configurable event settings.
 - Dashboard access is role-based.
 - Validation rules prevent inconsistent or duplicate data.
 - Audit logs capture administrative activities.
 - The platform remains stable under expected concurrent load.
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